**CS 409 : Introduction to Cryptography** Fall 2025

# Assignment 4

## One Way Functions, PKE, Key Exchange

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## Problem 1: It's a one-way street!

Let $F = \{f_k : \{0,1\}^n \to \{0,1\}^n\}_{k \in \{0,1\}^n}$ be a length-preserving pseudorandom permutation (c.f. Lec 8). Are the following functions one-way (c.f. Def 1, Lec 13 (same definition works for single function))? If yes, prove it via a reduction; else show an attack.

(a) $f(x, y) := f_x(y)$

(b) $f(y) := f_{0^n}(y)$ (where $n = |y|$).

(c) $f(x) := f_x(0^n)$ (where $n = |x|$).

(d) $f(x) := g(g(x))$ for one way function $g$.

(e) $f(x) := g(x)||g(g(x))$ for one way function $g$.

(f) $f$ is a one-to-one function which has a hard-core predicate $hc$ (c.f. Def 2, Lec 13).

## Problem 2: Easy or Hard?

a) Can the following problem be solved in polynomial time? Given a prime $p$, a value $x \in \mathbb{Z}_p^*$, and $y := g^x \mod p$, where $g$ is a uniform value in $\mathbb{Z}_p^*$. Find $g$, i.e., compute $y^{1/x} \mod p$. If yes, give a poly-time algorithm; else, show a reduction to any of the assumptions from class.

b) Show that the $\mathsf{DDH}$ problem (c.f. Lec 12) is easy in $\mathbb{Z}_p^*$. To solve this exercise, you'll need to know a few facts about $\mathbb{Z}_p^*$:

- An element $g \in \mathbb{Z}_p^*$ is a *quadratic residue* if it is a *square* modulo $p$: i.e., there exists $r \in \mathbb{Z}_p^*$ such that $r^2 = g \mod p$.
- Exactly *half* the elements in $\mathbb{Z}_p^*$ are quadratic residues (recall that $\mathbb{Z}_p^*$ has order $p-1$ and $p-1$ is even).
- There is an efficient algorithm to *test* whether or not $g \in \mathbb{Z}_p^*$ is a quadratic residue.

c) Let $\mathcal{G}$ be a polynomial-time algorithm that, on input $1^n$, outputs a prime $p$ with $\|p\| = n$ and a generator $g$ of $\mathbb{Z}_p^*$. The discrete logarithm problem is believed to be hard for $\mathcal{G}$. This means that the function (family) $f_{p,g}$ where $f_{p,g}(x) := [g^x \pmod p]$ is one-way. Let $\mathrm{lsb}(x)$ denote the least-significant bit of $x$. Show that lsb is not a hard-core predicate for $f_{p,g}$.

## Problem 3: Key Exchange

a) A group $\mathbb{G}$ with a group generation algorithm $\mathcal{G}$ is said to be a *gap* DH group if $\mathsf{DDH}$ is easy but $\mathsf{CDH}$ (c.f. Lec 12) is hard over $\mathbb{G}$ (example of such a group is $\mathbb{Z}_p^*$). Show that the Diffie-Hellman Key Exchange (c.f. Lec 14) is insecure when instantiated in a gap DH group.

b) Consider the following key exchange protocol:

- Alice chooses uniform $k, r \in \{0,1\}^n$, and sends $s := k \oplus r$ to Bob.
- Bob chooses uniform $t \in \{0,1\}^n$, and sends $u := s \oplus t$ to Alice.
- Alice computes $w := u \oplus r$ and sends $w$ to Bob.
- Alice outputs $k$ and Bob outputs $w \oplus t$.

Show that Alice and Bob output the same key. Analyze the security of the scheme (i.e., either prove its security or show a concrete attack).

# Problem 4: Public Speaking

a) Show that any two-round key-exchange protocol (c.f. Def 1, Lec 14), where Alice sends one message and Bob replies with another message before they output the keys, can be converted into an IND-CPA secure public key encryption scheme (c.f. Lec 14).

b) Consider the following variant of ElGamal encryption ((c.f. Lec 14). Let $p = 2q + 1$, let $\mathbb{G}$ be the group of squares modulo $p$ (so $\mathbb{G}$ is a subgroup of $\mathbb{Z}_p^*$ of order $q$), and let $g$ be a generator of $\mathbb{G}$. The private key is $(\mathbb{G}, g, q, x)$ and the public key is $(\mathbb{G}, g, q, h)$ where $h = g^x$ and $x \in \mathbb{Z}_q$, choose a uniform $r \in \mathbb{Z}_q$, compute $c_1 := g^r \pmod p$ and $c_2 := h^r + m \pmod p$, and let the ciphertext be $\langle c_1, c_2 \rangle$. Is this scheme CPA-secure? Prove your answer.

c) Let $\Pi_{bit} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ be an IND-CPA secure PKE for $\mathcal{M}_{bit} = \{0,1\}$ (single bit). Can you construct a $\Pi'$ using $\Pi$ such that $\Pi'$ is IND-CPA secure PKE for $\mathcal{M} = \{0,1\}^n$ ($n$-bit strings). Describe the construction and give a proof of security via a hybrid argument.

d) Consider *one-wayness* under CPA (OW-CPA), an alternative security notion for PKE defined as follows:

- $\mathsf{Eve}$ is given $pk$, generated as $(pk, sk) \leftarrow \mathsf{Gen}(1^n)$.
- For $m \leftarrow \mathcal{M}_n$, $\mathsf{Eve}$ is given $c \leftarrow \mathsf{Enc}(pk, m)$ as the challenge ciphertext.
- $\mathsf{Eve}$ outputs $m'$ and breaks if $m' = m$.

A PKE $\Pi$ is OW-CPA-secure if for all PPT eavesdroppers $\mathsf{Eve}$, the probability with which $\mathsf{Eve}$ breaks $\Pi$ as above is negligible. Now answer the following questions about IND-CPA and OW-CPA.

i) Show formally that IND-CPA implies OW-CPA. That is, *any* PKE that is IND-CPA-secure is also OW-CPA-secure.

ii) What about the opposite direction? Show either that

   i. OW-CPA implies IND-CPA; or

   ii. Come up with a counterexample, i.e., a PKE $\Pi$ that is OW-CPA-secure but not IND-CPA-secure.